Paper G-03: AI Emergence: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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May 2026

Abstract

This paper develops a falsifiable CDFD/AFL model of AI Emergence. It maps data, compute, models, feedback, deployment demand, and social attention to driving flux Φ, compute, data quality, governance, evaluation, energy, and oversight to active constraint C, learning, tool use, modularity, monitoring, correction, and institutional response to responsiveness S, and weights, deployed systems, standards, incidents, incentives, and feedback history to structural memory M_s. The target outcome is capability, reliability, diffusion, risk, and social adaptation, tested against scaling records, evaluations, deployment telemetry, incidents, policy changes, and energy use. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow–constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine

[5]. Part IV [6], DOI `https://doi.org/10.5281/zenodo.20395901`, is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [3], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \to \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

Artificial neural networks simulate learning by iteratively adjusting millions or billions of parameters to minimize a loss function against a training dataset. The current paradigm of AI development relies on massive brute-force scaling: pumping enormous volumes of data through immense clusters of GPUs.

While this has yielded emergent cognitive behaviors, the trajectory is fundamentally thermodynamic. The network is essentially an active engine consuming information and energy to produce structured mathematical memory. In CDFD terms, a deep learning model is an adaptive topological surface attempting to balance immense throughput against strict physical and architectural bottlenecks.

3 Neural Architecture as an Adaptive Surface

We map the training of a neural network to the CDFD life number equation:

$$\Psi_s = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s \tag{1}$$

In an artificial intelligence system:

- **Flux (Φ)**: The gradient updates and the raw throughput of training tokens (data flow).
- **Constraint (C)**: The rigid, fixed architecture of the network (e.g., fixed number of layers and attention heads), as well as the physical hardware limits (memory bandwidth and thermal dissipation).
- **Responsiveness (S)**: The available parameter space (model width and depth) that can actively adapt to the incoming gradients.

- **Memory (M_s)**: The accumulated, frozen weights of the network. As the model trains, it transitions from a high-entropy chaotic state into a low-entropy locked state (overfitting or saturation).

3.1 The Saturation Threshold and Catastrophic Forgetting

During early training, S is high (many free parameters) and M_s is low. The network efficiently absorbs the flux Φ. $\Psi_s \approx 1$.

However, as training progresses, the weights solidify into fixed representations. The systemic memory M_s increases, locking the topology. If new, out-of-distribution data (Φ) is introduced, the fixed architectural constraints (C) cannot flex to accommodate it. The network enters an over-constrained regime ($\Psi_s \ll 1$). To learn the new data, it must overwrite its existing weights, destroying its previous capabilities—a phenomenon known in machine learning as "catastrophic forgetting."

Unlike biological brains, which dynamically grow new synapses and prune old ones (active restructuring of C and S), current LLMs have fixed, static constraints.

4 The Path to General Intelligence

The CDFD framework models the condition that that you cannot achieve infinite adaptability in a system with fixed static constraints. If C is locked, Ψ_s will enter an overload regime as M_s saturates.

True Artificial General Intelligence (AGI) requires an architecture where the constraint C itself is a dynamic variable governed by the differential equation $\frac{\partial C}{\partial t} = \alpha|\Phi| - \beta C$. The network must be able to physically or logically grow new architectural branches (neurogenesis) and prune useless nodes to continually maintain $\Psi_s \approx 1$. Liquid neural networks and sparsely activated mixture-of-experts (MoE) are primitive steps toward this, but an AGI must be a fully self-regulating CDFD surface.

5 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

5.1 Current Runtime Diagnostic: Universal Network Cascade

The release-local discovery script keeps this paper tied to the current CDFD Runtime [7], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the 2026-06-04 runtime pass, the localized hub-drive stress test ended with hub

$\Psi_s = 5.21$, peak network $\Psi_s = 5.21$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

6 Major Universal Upgrade: Mechanism, Scale, and Tests

6.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For AI Emergence, the proposed drive is data, compute, models, feedback, deployment demand, and social attention. The active limitation is compute, data quality, governance, evaluation, energy, and oversight. Responsiveness is carried by learning, tool use, modularity, monitoring, correction, and institutional response, while retained state is represented by weights, deployed systems, standards, incidents, incentives, and feedback history. The outcome to explain is capability, reliability, diffusion, risk, and social adaptation.

Table 1: Operational CDFD/AFL map for Paper G-03.

Term	**Paper-specific observable meaning**	**Measurement question**
Φ	data, compute, models, feedback, deployment demand, and social attention	What rate, load, gradient, or demand enters the focal system?
C	compute, data quality, governance, evaluation, energy, and oversight	Which measured bottleneck limits transfer, function, or recovery?
S	learning, tool use, modularity, monitoring, correction, and institutional response	Which process changes effective capacity or reroutes the load?
M_s	weights, deployed systems, standards, incidents, incentives, and feedback history	Which retained state makes equal present loads produce different futures?
Y	capability, reliability, diffusion, risk, and social adaptation	Which outcome is predicted out of sample, with uncertainty?

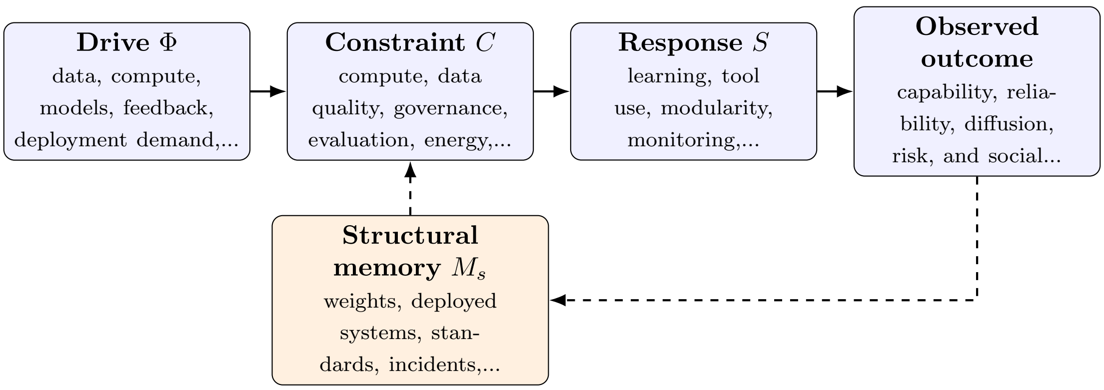

Figure 1: Paper G-03 observable CDFD/AFL architecture for AI Emergence. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

6.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \tag{2}$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \tag{3}$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \tag{4}$$

$$\Psi_s(t) = \frac{\widehat{\Phi}(t)}{\epsilon + \widehat{C}(t)}\,\widehat{S}(t)\,[1 + \omega\widehat{M_s}(t)]. \tag{5}$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in data, compute, models, feedback, deployment demand, and social attention arrives faster than learning, tool use, modularity, monitoring, correction, and institutional response can compensate. This raises or redistributes compute, data quality, governance, evaluation, energy, and oversight. If the load persists, the retained state encoded by weights, deployed systems, standards, incidents, incentives, and feedback history changes the next response, so a later event of equal magnitude can produce a different trajectory in capability, reliability, diffusion, risk, and social adaptation. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

6.3 Cross-Scale Universality Without Mechanistic Erasure

For the abstract and cognitive systems family, the universal comparison is between information-bearing activity, limited channels or institutions, adaptive reorganization, and retained semantic or technical structure. Because meanings and goals are not conserved physical quantities, every mapping must state its observational unit and avoid treating metaphor as measurement. [8, 2, 10, 9, 1] The CDFD programme is cumulative: Part I supplies the flow–constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is microseconds to decades; model components to socio-technical ecosystems. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

6.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper G-03. Each rung can fail independently.

Test	**Design**	**Discriminating result**
Proxy audit	Measure scaling records, evaluations, deployment telemetry, incidents, policy changes, and energy use and preregister how each record maps to Φ, C, S, M_s, and Y.	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a capability or deployment shock under contrasting oversight histories while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper's stated falsifier is: CDFD variables do not improve capability, failure, or governance-response prediction.

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ, separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

6.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from scaling records, evaluations, deployment telemetry, incidents, policy changes, and energy use and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of capability, reliability, diffusion, risk, and social adaptation. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

7 Conclusion

Artificial-intelligence systems face measurable limits in compute, data, energy, evaluation, and governance. CDFD offers a candidate architecture for testing how those constraints interact with learning and deployment history; it does not establish a universal ceiling or guarantee open-ended cognition.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

8 Reference Layer

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